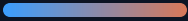


Enterprise Adoption & Escalation Risk Analyzer



One explainable **0–100 escalation-risk score** per customer account — fused from five systems, broken down factor-by-factor, and turned into a **Claude-written action plan**.

THE PROBLEM

Account health hides in five different systems.



Support

Cases, severities, escalations, aging tickets



Infrastructure

Cluster health, alerts, end-of-life exposure



Customer Success

Adoption, health score, sentiment



Renewals

Contract status & renewal proximity



Engagement

Internal touchpoints & escalation chatter



The result

No single number. Escalations surface too late. Teams triage by gut, not signal.

THE SOLUTION

One number. Fully explainable. Immediately actionable.



Unified risk score

A weighted engine merges ~15 signals into one 0–100 escalation-risk score per account.



Full explainability

Every score decomposes into ranked factors with each source's contribution.



Claude action plan

A server-side Claude advisory turns the profile into a prioritized weekly plan.

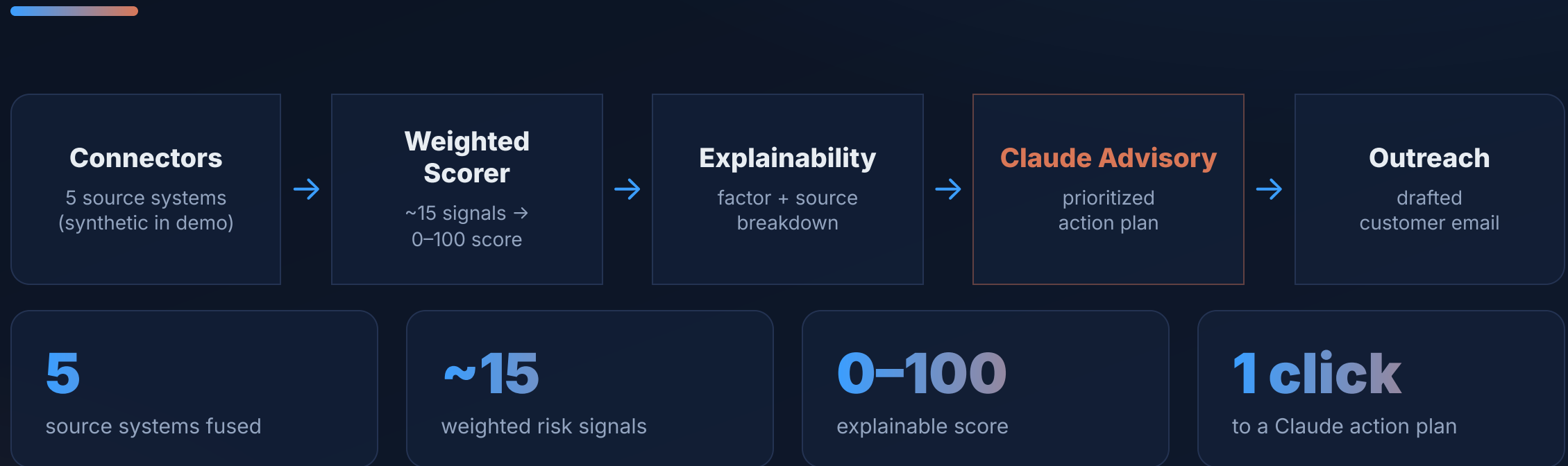


Auto-drafted outreach

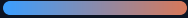
Generates the customer email with flagged problems and next steps.

HOW IT WORKS

From scattered signals to a scored, explained, actionable account.



No black-box scores.



Every account's score breaks down into ranked, weighted factors — so a Customer Experience Manager sees **exactly why** an account is at risk and where to look first.

- Score-by-source donut + every weighted factor, charted — support / infra / CS / engagement / manual flags
- Transparent weights, tunable in one file — P1/P2 ratio and escalation history carry the most
- Manual CXM risk flags layer on top — human judgment + the model, not either alone

A real Claude integration — built responsibly.



Server-side only

The Anthropic call runs in the backend; the API key never reaches the browser.



Explicit & cached

Claude is invoked only on a click, and cached per account — cost stays negligible.



Zero-cost fallback

With no key, a deterministic advisory is synthesized locally — it never breaks.



Grounded

The prompt carries only computed signals — no PII, no invented data.

Cost-aware, graceful, and honest — the engineering values that matter for production LLM systems.

ARCHITECTURE

A clean, deployable ASGI app.



Dashboard — HTML · vanilla JS · Chart.js

Risk table · Portfolio donut · Drill-downs · AI Advisory · Email composer

FastAPI backend

Aggregator · Risk Scorer · Claude Advisor · Email Generator · TTL cache

Connector layer (demo: synthetic generators)

support · infrastructure · CS platform · renewals · internal engagement

Persistence & jobs

SQLite · optional vector store · APScheduler background recompute

Full-stack applied AI, end to end.



Python

FastAPI

Pydantic v2

SQLite

APScheduler

Chart.js

Anthropic Claude API

Docker

Render

Data fusion

Merging messy, multi-system signals into one coherent model.

Explainable scoring

Transparent, weighted, factor-level — not a black box.

Safe LLM integration

Server-side, gated, fallback-backed Claude usage.

Full-stack delivery

API, UI, background jobs, persistence, charts.

Cloud deploy

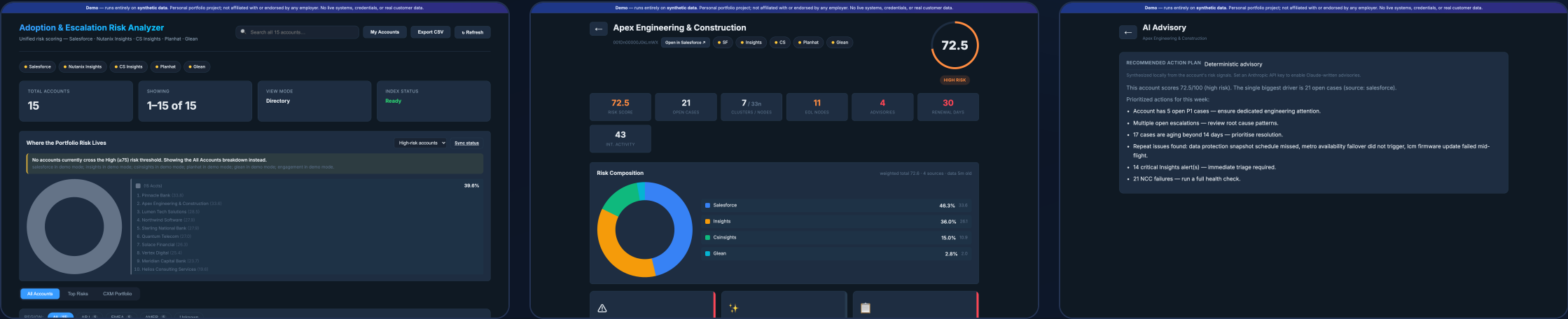
Dockerised, blueprint deploy, keep-alive automation.

Responsible defaults

Synthetic data, no secrets, honest disclaimers.

SEE IT LIVE

Click through the whole flow.



enterprise-adoption-risk-analyzer.onrender.com

Runs entirely on synthetic data — no real customers, credentials, or live systems.

ABOUT

Literally the Applied AI / Solutions Architect job — in code.



Aritra De — an enterprise technical, customer-facing operator (10+ years across VMware, Nutanix, Veeam) now building applied-AI systems for Indian BFSI & enterprise.

Live demo enterprise-adoption-risk-analyzer.onrender.com

Source github.com/aritrade/enterprise-adoption-risk-analyzer

Portfolio aritra-de.vercel.app